

Re-ERP Inventory Management Module

Requirements & Design Document (RD) - v1.0 - September 2026

Prepared for the Re-ERP product team. Defines the data model, web services and screens for the Inventory Management module, and the roadmap toward Purchasing, Order Management and Cost Accounting.

1. Executive Summary

This document specifies a complete Inventory Management module for Re-ERP. It defines the organization hierarchy (master and child inventory organizations with business-unit assignment, and cost organizations that group inventory organizations for valuation), organization-level parameter sets (inventory, receiving, shipping), subinventories and locators, the item master with per-organization assignments, the inventory transaction engine (receipts, issues, subinventory transfers), inter-organization transfers (direct and in-transit), on-hand balance maintenance, and the accounting hand-off into the existing Re-ERP General Ledger.

It contains the full database design (tables, columns, relationships), the complete ORDS web-service catalog with sample request/response payloads, and screen designs for every page, following the UI patterns already established across Re-ERP: Redwood styling, tabbed workspaces, LOV-driven forms, API-transparency dialogs, and the guarded SQL gateway for inquiries.

The module is Phase 1 of the supply-chain roadmap. Purchasing, Order Management, and Cost Accounting will be layered on top of these structures, so the data model reserves the linkage points those modules will need (transaction source/document references, cost organization books, receiving tolerances).

1.1 Scope of this phase

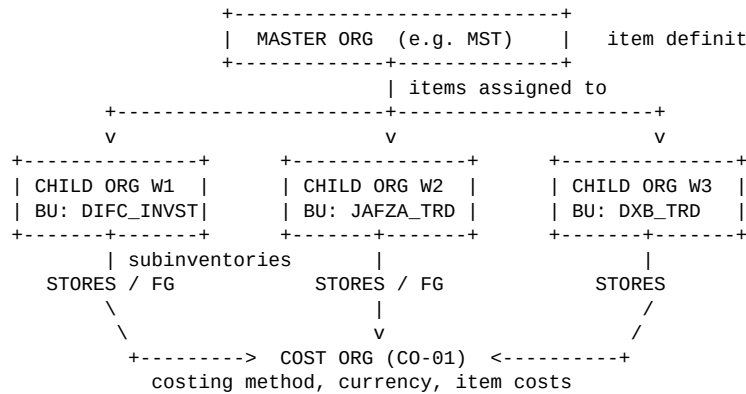
- Inventory organization setup: one master organization per company, any number of child organizations
- Child organizations assigned to a Business Unit (existing gl/businessunits data) and its legal entity
- Organization parameter sets: inventory parameters, receiving parameters, shipping parameters
- Subinventories per organization, optional locators within a subinventory
- Cost organizations; each inventory organization assigned to exactly one cost organization
- Item master: items created in the master organization, then assigned to child organizations
- Units of measure master (seeded)
- Transactions inside an organization: miscellaneous receipt, miscellaneous issue, subinventory transfer
- Inter-organization transfers: direct (one step) and in-transit (ship, then receive)
- On-hand balances by organization / subinventory / locator / item, with inquiry page
- GL accounting distributions for every material transaction, posted through the existing journal interface
- Item cost per cost organization (simple current cost) so every transaction can be valued

1.2 Out of scope (later phases)

- Purchasing: requisitions, purchase orders, PO receiving against orders - Phase 2
- Order Management: sales orders, picking, shipping against orders - Phase 3
- Cost Accounting: perpetual average recalculation, standard cost updates, cost history and reports - Phase 4
- Lot and serial control (control flags are reserved in the schema; validation logic later)
- Min-max planning, replenishment, cycle counting, physical inventory

2. Organization Model

A single organization per company is flagged as the ITEM MASTER organization: items are defined once there. Child organizations represent physical warehouses or plants; each is linked to its master organization, assigned to a Business Unit, and inherits that BU's legal entity. An item becomes usable in a child organization only after an explicit item-organization assignment. Cost organizations sit alongside the inventory hierarchy: every inventory organization belongs to exactly one cost organization, which carries the costing method, the costing currency, and the item costs used to value transactions.



Rules enforced by the services: a child organization must reference an existing master organization and a Business Unit; an organization cannot be deactivated while it holds non-zero on-hand; a master organization cannot be deleted while items exist.

3. Data Model

3.1 Relationships at a glance

Parent	Child	Link	Cardinality
RR_INV_ORGANIZATIONS (master)	RR_INV_ORGANIZATIONS (child)	MASTER_ORG_ID	1 : N
GL business units (existing)	RR_INV_ORGANIZATIONS	BUSINESS_UNIT_NAME	1 : N
RR_INV_ORGANIZATIONS	RR_INV_ORG_PARAMETERS	ORGANIZATION_ID	1 : 1
RR_INV_ORGANIZATIONS	RR_INV_RECEIVING_PARAMETERS	ORGANIZATION_ID	1 : 1
RR_INV_ORGANIZATIONS	RR_INV_SHIPPING_PARAMETERS	ORGANIZATION_ID	1 : 1
RR_INV_ORGANIZATIONS	RR_INV_SUBINVENTORIES	ORGANIZATION_ID	1 : N
RR_INV_SUBINVENTORIES	RR_INV_LOCATORS	SUBINVENTORY_ID	1 : N
RR_INV_COST_ORGANIZATIONS	RR_INV_COST_ORG_ASSIGNMENTS	COST_ORG_ID	1 : N
RR_INV_ORGANIZATIONS	RR_INV_COST_ORG_ASSIGNMENTS	ORGANIZATION_ID	1 : 1
RR_INV_ORGANIZATIONS (master)	RR_INV_ITEMS	MASTER_ORG_ID	1 : N
RR_INV_ITEMS	RR_INV_ITEM_ORG_ASSIGNMENTS	ITEM_ID	1 : N
RR_INV_ORGANIZATIONS (child)	RR_INV_ITEM_ORG_ASSIGNMENTS	ORGANIZATION_ID	1 : N
RR_INV_UOMS	RR_INV_ITEMS	PRIMARY_UOM_CODE	1 : N
RR_INV_TXN_TYPES	RR_INV_TRANSACTIONS	TXN_TYPE_CODE	1 : N
RR_INV_ORGANIZATIONS	RR_INV_TRANSACTIONS	ORGANIZATION_ID	1 : N
RR_INV_ITEMS	RR_INV_TRANSACTIONS	ITEM_ID	1 : N
RR_INV_INTERORG_SHIPMENTS	RR_INV_TRANSACTIONS	SHIPMENT_ID	1 : N
org + subinv + locator + item	RR_INV_ONHAND_BALANCES	composite unique	1 row per combination
RR_INV_COST_ORGANIZATIONS	RR_INV_ITEM_COSTS	COST_ORG_ID	1 : N
RR_INV_TRANSACTIONS	RR_INV_TXN_ACCOUNTING	TRANSACTION_ID	1 : N (Dr and Cr lines)

ERD (simplified)

```

GL_BUSINESS_UNITS 1--N RR_INV_ORGANIZATIONS 1--1 RR_INV_ORG_PARAMETERS
| 1--1 RR_INV_RECEIVING_PARAMETERS
| 1--1 RR_INV_SHIPPING_PARAMETERS
| 1--N RR_INV_SUBINVENTORIES 1--N RR_INV_LOCATORS
| 1--N RR_INV_ITEM_ORG_ASSIGNMENTS N--1 RR_INV_ITEMS N--1 RR_INV_UOMS
| 1--N RR_INV_TRANSACTIONS N--1 RR_INV_TXN_TYPES
| 1--1 RR_INV_COST_ORG_ASSIGNMENTS N--1 RR_INV_COST_ORGANIZATIONS
| 1--N RR_INV_ONHAND_BALANCES | 1--N RR_INV_ITEM_COSTS
RR_INV_INTERORG_SHIPMENTS 1--N RR_INV_TRANSACTIONS 1--N RR_INV_TXN_ACCOUNTING
  
```

All primary keys come from one sequence RR_INV_SEQ starting at 5000000001, so ids never collide with data synced from external systems.

3.2 RR_INV_ORGANIZATIONS

Column	Type	Req	Description
ORGANIZATION_ID	NUMBER	Y	PK (RR_INV_SEQ)
ORGANIZATION_CODE	VARCHAR2(18)	Y	Short unique code (MST, W1, W2 ...)
ORGANIZATION_NAME	VARCHAR2(240)	Y	Display name
ORG_TYPE	VARCHAR2(10)	Y	MASTER or CHILD
MASTER_ORG_ID	NUMBER	child	FK to the master org; NULL on a master row
BUSINESS_UNIT_NAME	VARCHAR2(240)	child	Assigned BU (from gl/businessunits)
LEGAL_ENTITY_NAME	VARCHAR2(360)	N	Copied from the BU at assignment time
LOCATION	VARCHAR2(240)	N	Site / address free text
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE (default ACTIVE)
CREATED_BY, CREATION_DATE	-	Y	Audit
LAST_UPDATED_BY, LAST_UPDATE_DATE	-	Y	Audit

Constraints: UNIQUE(ORGANIZATION_CODE); CHECK(ORG_TYPE IN ('MASTER','CHILD')). The create service enforces that a CHILD row carries MASTER_ORG_ID and BUSINESS_UNIT_NAME.

3.3 RR_INV_ORG_PARAMETERS (1:1 with organization)

Column	Type	Req	Description
ORGANIZATION_ID	NUMBER	Y	PK and FK
PRIMARY_UOM_CODE	VARCHAR2(10)	Y	Default UOM (EA)
DEFAULT_SUBINVENTORY	VARCHAR2(30)	N	Default receiving subinventory
LOCATOR_CONTROL	VARCHAR2(15)	Y	NONE / PRESPECIFIED (default NONE)
NEGATIVE_INV_ALLOWED_FLAG	VARCHAR2(1)	Y	Y/N - allow negative on-hand (default N)
LOT_CONTROL_ENABLED_FLAG	VARCHAR2(1)	Y	Reserved (default N)
SERIAL_CONTROL_ENABLED_FLAG	VARCHAR2(1)	Y	Reserved (default N)
CURRENCY_CODE	VARCHAR2(15)	Y	Valuation currency (default AED)
MATERIAL_ACCOUNT	VARCHAR2(500)	Y	Code combination: inventory valuation
ADJUSTMENT_ACCOUNT	VARCHAR2(500)	Y	Offset for miscellaneous receipt/issue
INTERORG_RECEIVABLE_ACCOUNT	VARCHAR2(500)	N	Inter-org receivable
INTERORG_PAYABLE_ACCOUNT	VARCHAR2(500)	N	Inter-org payable
INTRANSIT_ACCOUNT	VARCHAR2(500)	N	In-transit inventory
Audit columns	-	Y	-

Account fields store full code combinations chosen with the existing AccountSelector component, with the company segment locked to the BU's company code (same pattern as Manage Receipts and the bank-account setup).

3.4 RR_INV_RECEIVING_PARAMETERS (1:1 with organization)

Column	Type	Req	Description
ORGANIZATION_ID	NUMBER	Y	PK and FK
RECEIPT_ROUTING	VARCHAR2(15)	Y	DIRECT / STANDARD / INSPECTION (default DIRECT)
RECEIPT_NUM_GENERATION	VARCHAR2(10)	Y	AUTOMATIC / MANUAL (default AUTOMATIC)
NEXT_RECEIPT_NUMBER	NUMBER	Y	Counter used when AUTOMATIC (default 1)
OVER_RECEIPT_TOLERANCE_PERCENT	NUMBER	N	Percent over ordered qty allowed (used by Phase 2)
OVER_RECEIPT_ACTION	VARCHAR2(10)	N	NONE / WARNING / REJECT (Phase 2)
EARLY_RECEIPT_DAYS / LATE_RECEIPT_DAYS	NUMBER	N	Date tolerances (Phase 2)

Column	Type	Req	Description
ALLOW_SUBSTITUTE_RECEIPTS_FLAG	VARCHAR2(1)	N	Y/N (Phase 2)
RECEIVING_INVENTORY_ACCOUNT	VARCHAR2(500)	N	Receiving clearing account
Audit columns	-	Y	-

3.5 RR_INV_SHIPPING_PARAMETERS (1:1 with organization)

Column	Type	Req	Description
ORGANIZATION_ID	NUMBER	Y	PK and FK
SHIPMENT_NUM_GENERATION	VARCHAR2(10)	Y	AUTOMATIC / MANUAL (default AUTOMATIC)
NEXT_SHIPMENT_NUMBER	NUMBER	Y	Counter (default 1)
DEFAULT_SHIP_METHOD	VARCHAR2(60)	N	Carrier / method free text
PICK_CONFIRM_REQUIRED_FLAG	VARCHAR2(1)	N	Y/N (Phase 3)
GOODS_IN_TRANSIT_ACCOUNT	VARCHAR2(500)	N	Used by in-transit transfers
Audit columns	-	Y	-

3.6 RR_INV_SUBINVENTORIES

Column	Type	Req	Description
SUBINVENTORY_ID	NUMBER	Y	PK
ORGANIZATION_ID	NUMBER	Y	FK
SUBINVENTORY_CODE	VARCHAR2(30)	Y	Unique within the org (STORES, FG, DAMAGED ...)
DESCRIPTION	VARCHAR2(240)	N	-
SUBINV_TYPE	VARCHAR2(15)	Y	STORAGE / RECEIVING (default STORAGE)
LOCATOR_CONTROL	VARCHAR2(15)	Y	NONE / PRESPECIFIED (defaults from org)
MATERIAL_ACCOUNT	VARCHAR2(500)	N	Overrides org material account when set
ASSET_FLAG	VARCHAR2(1)	Y	Y = valued stock, N = expense subinventory
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE
Audit columns	-	Y	-

Constraint: UNIQUE(ORGANIZATION_ID, SUBINVENTORY_CODE). A subinventory with on-hand cannot be deactivated.

3.7 RR_INV_LOCATORS

Column	Type	Req	Description
LOCATOR_ID	NUMBER	Y	PK
SUBINVENTORY_ID	NUMBER	Y	FK
LOCATOR_CODE	VARCHAR2(60)	Y	e.g. A1-R2-B3; unique within the subinventory
DESCRIPTION	VARCHAR2(240)	N	-
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE

3.8 RR_INV_UOMS

Column	Type	Req	Description
UOM_CODE	VARCHAR2(10)	Y	PK (EA, KG, G, M, CM, L, ML, BOX, PKT, ROLL)
UOM_NAME	VARCHAR2(60)	Y	Each, Kilogram, ...
UOM_CLASS	VARCHAR2(30)	Y	QUANTITY / WEIGHT / LENGTH / VOLUME
BASE_FLAG	VARCHAR2(1)	Y	Y = base unit of its class
CONVERSION_TO_BASE	NUMBER	N	Multiplier to class base (conversions in a later phase)

3.9 RR_INV_ITEMS (item master)

Column	Type	Req	Description
ITEM_ID	NUMBER	Y	PK
MASTER_ORG_ID	NUMBER	Y	FK to the master organization
ITEM_NUMBER	VARCHAR2(80)	Y	Unique item code
DESCRIPTION	VARCHAR2(240)	Y	-
LONG_DESCRIPTION	VARCHAR2(2000)	N	-
ITEM_TYPE	VARCHAR2(30)	Y	FINISHED_GOOD / PURCHASED / RAW_MATERIAL / SERVICE / OTHER
CATEGORY	VARCHAR2(80)	N	Free category (category sets in a later phase)
PRIMARY_UOM_CODE	VARCHAR2(10)	Y	FK to RR_INV_UOMS
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE
STOCKABLE_FLAG	VARCHAR2(1)	Y	Default Y
TRANSACTABLE_FLAG	VARCHAR2(1)	Y	Default Y
PURCHASABLE_FLAG	VARCHAR2(1)	Y	Used by Phase 2 Purchasing
SALEABLE_FLAG	VARCHAR2(1)	Y	Used by Phase 3 Order Management

Column	Type	Req	Description
LOT_CONTROL_FLAG / SERIAL_CONTROL_FLAG	VARCHAR2(1)	Y	Reserved (default N)
LIST_PRICE	NUMBER	N	Default selling price
STANDARD_COST	NUMBER	N	Fallback cost when no item-cost row exists
ATTRIBUTE1..ATTRIBUTE5	VARCHAR2(240)	N	Spare flexfield-style attributes
Audit columns	-	Y	-

3.10 RR_INV_ITEM_ORG_ASSIGNMENTS

Column	Type	Req	Description
ASSIGNMENT_ID	NUMBER	Y	PK
ITEM_ID	NUMBER	Y	FK
ORGANIZATION_ID	NUMBER	Y	FK to a CHILD organization
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE
DEFAULT_SUBINVENTORY	VARCHAR2(30)	N	Default for this item in this org
MIN_QTY / MAX_QTY	NUMBER	N	Reserved for min-max planning
Audit columns	-	Y	-

Constraint: UNIQUE(ITEM_ID, ORGANIZATION_ID). The transaction engine rejects any movement of an item that has no ACTIVE assignment in that organization.

3.11 RR_INV_COST_ORGANIZATIONS and RR_INV_COST_ORG_ASSIGNMENTS

Column	Type	Req	Description
COST_ORG_ID	NUMBER	Y	PK
COST_ORG_CODE	VARCHAR2(30)	Y	Unique
COST_ORG_NAME	VARCHAR2(240)	Y	-
COST_METHOD	VARCHAR2(15)	Y	AVERAGE / STANDARD (applies to assigned orgs)
CURRENCY_CODE	VARCHAR2(15)	Y	Costing currency
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE
Audit columns	-	Y	-

RR_INV_COST_ORG_ASSIGNMENTS: ASSIGNMENT_ID (PK), COST_ORG_ID (FK), ORGANIZATION_ID (FK, UNIQUE - an inventory organization belongs to at most one cost organization), audit columns. Transactions resolve their unit cost through this assignment.

3.12 RR_INV_ITEM_COSTS

Column	Type	Req	Description
ITEM_COST_ID	NUMBER	Y	PK
COST_ORG_ID	NUMBER	Y	FK
ITEM_ID	NUMBER	Y	FK
COST_PER_UNIT	NUMBER	Y	Current unit cost in the cost-org currency
EFFECTIVE_FROM	DATE	Y	-
Audit columns	-	Y	-

Phase 1 keeps one current cost per item per cost organization, maintained manually on the Cost Organizations page or set by the first receipt. Phase 4 replaces manual maintenance with perpetual average recalculation and full cost history.

3.13 RR_INV_TXN_TYPES (seeded)

TXN_TYPE_CODE	Name	Direction	Accounting (Dr / Cr)
MISC_RECEIPT	Miscellaneous Receipt	+ into subinventory	Material account / Adjustment account
MISC_ISSUE	Miscellaneous Issue	- from subinventory	Adjustment account / Material account
SUBINV_TRANSFER	Subinventory Transfer	- from, + to (same org)	Material(to) / Material(from); net zero when accounts match
INTERORG_DIRECT	Direct Inter-Org Transfer	- source org, + destination org	See section 5.1
INTERORG_SHIP	In-Transit Shipment	- source org, + in-transit	See section 5.2
INTERORG_RECEIVE	In-Transit Receipt	- in-transit, + destination org	See section 5.2
PO_RECEIPT	PO Receipt (Phase 2)	+	Reserved
RETURN_TO_VENDOR	Return To Vendor (Phase 2)	-	Reserved
SO_ISSUE	Sales Order Issue (Phase 3)	-	Reserved

TXN_TYPE_CODE	Name	Direction	Accounting (Dr / Cr)
COST_ADJ	Cost Adjustment (Phase 4)	0 qty	Reserved

3.14 RR_INV_TRANSACTIONS

One row per material movement leg. A subinventory transfer writes two rows (out and in) sharing a TRANSFER_GROUP_ID; an inter-org transfer writes one or two rows per organization linked by SHIPMENT_ID.

Column	Type	Req	Description
TRANSACTION_ID	NUMBER	Y	PK
ORGANIZATION_ID	NUMBER	Y	Org where the movement happens
TXN_TYPE_CODE	VARCHAR2(30)	Y	FK to RR_INV_TXN_TYPES
ITEM_ID	NUMBER	Y	FK
SUBINVENTORY_CODE	VARCHAR2(30)	Y	Subinventory of this leg
LOCATOR_ID	NUMBER	N	When locator controlled
QUANTITY	NUMBER	Y	Signed: + receipt, - issue
UOM_CODE	VARCHAR2(10)	Y	Transaction UOM (primary UOM in Phase 1)
UNIT_COST	NUMBER	Y	Cost applied to this leg
TXN_VALUE	NUMBER	Y	QUANTITY x UNIT_COST (signed)
TRANSACTION_DATE	DATE	Y	Accounting date
TRANSFER_GROUP_ID	NUMBER	N	Groups the two legs of a subinventory transfer
TRANSFER_ORG_ID	NUMBER	N	Counterpart org on inter-org movements
TRANSFER_SUBINVENTORY	VARCHAR2(30)	N	Counterpart subinventory
SHIPMENT_ID	NUMBER	N	FK to RR_INV_INTERORG_SHIPMENTS
SOURCE_TYPE	VARCHAR2(30)	Y	INV (this phase); PO / OM / COST reserved
SOURCE_REFERENCE	VARCHAR2(80)	N	Free reference / document number
REASON	VARCHAR2(240)	N	Reason / comment
ACCOUNTED_FLAG	VARCHAR2(1)	Y	N until GL distributions are posted
JE_BATCH_ID	NUMBER	N	GL batch created for this transaction
Audit columns	-	Y	-

3.15 RR_INV_ONHAND_BALANCES

Column	Type	Req	Description
ONHAND_ID	NUMBER	Y	PK
ORGANIZATION_ID	NUMBER	Y	-
SUBINVENTORY_CODE	VARCHAR2(30)	Y	-
LOCATOR_ID	NUMBER	N	NULL when not locator controlled
ITEM_ID	NUMBER	Y	-
ONHAND_QTY	NUMBER	Y	Current quantity in primary UOM
LAST_TXN_ID	NUMBER	N	Last transaction that touched this row
LAST_UPDATE_DATE	TIMESTAMP	Y	-

Constraint: UNIQUE(ORGANIZATION_ID, SUBINVENTORY_CODE, NVL(LOCATOR_ID,0), ITEM_ID). Updated in the same database transaction as the movement row, so on-hand and history can never diverge.

3.16 RR_INV_INTERORG_SHIPMENTS

Column	Type	Req	Description
SHIPMENT_ID	NUMBER	Y	PK
SHIPMENT_NUMBER	VARCHAR2(30)	Y	From shipping parameters numbering
FROM_ORG_ID / TO_ORG_ID	NUMBER	Y	Source and destination orgs

Column	Type	Req	Description
TRANSFER_MODE	VARCHAR2(10)	Y	DIRECT / INTRANSIT
STATUS	VARCHAR2(12)	Y	SHIPPED / RECEIVED / (DIRECT rows are created RECEIVED)
SHIP_DATE / RECEIVE_DATE	DATE	Y/N	-
SHIP_METHOD	VARCHAR2(60)	N	-
NOTES	VARCHAR2(500)	N	-
Audit columns	-	Y	-

Lines are the RR_INV_TRANSACTIONS rows carrying this SHIPMENT_ID, so a shipment can move multiple items.

3.17 RR_INV_TXN_ACCOUNTING

Column	Type	Req	Description
ACCT_LINE_ID	NUMBER	Y	PK
TRANSACTION_ID	NUMBER	Y	FK
LINE_TYPE	VARCHAR2(2)	Y	DR / CR
ACCOUNT_COMBINATION	VARCHAR2(500)	Y	Full code combination
AMOUNT	NUMBER	Y	Absolute amount
CURRENCY_CODE	VARCHAR2(15)	Y	-
POSTED_FLAG	VARCHAR2(1)	Y	N until included in a GL batch
JE_BATCH_ID	NUMBER	N	-

4. Transaction Engine

4.1 Processing algorithm (single service, all types)

```
POST inv/transactions/create
1 Validate organization is ACTIVE and item has an ACTIVE assignment in it
2 Validate subinventory (and locator when controlled) exists and is ACTIVE
3 Resolve unit cost:
  cost org of the organization -> RR_INV_ITEM_COSTS.COST_PER_UNIT
  fallback -> RR_INV_ITEMS.STANDARD_COST -> else reject with 409
4 For negative-direction legs: check resulting on-hand >= 0
  unless org parameter NEGATIVE_INV_ALLOWED_FLAG = 'Y'
5 Insert RR_INV_TRANSACTIONS row(s) (two legs for transfers)
6 MERGE into RR_INV_ONHAND_BALANCES (same DB transaction)
7 Build RR_INV_TXN_ACCOUNTING Dr/Cr lines from the type rules (3.13)
8 COMMIT and return the new transaction id(s) and updated on-hand
9 GL posting runs as a separate action (4.3), like Create Accounting elsewhere in Re-ERP
```

4.2 Validation rules

- Quantity must be > 0 in the payload; the engine applies the sign from the transaction type
- Transaction date must fall in an open GL period (checked against the existing periods data)
- Subinventory transfers require different from/to subinventories in the same organization
- Inter-org movements require an ACTIVE item assignment in BOTH organizations
- An org with LOCATOR_CONTROL = PRESPECIFIED requires a locator on every leg in controlled subinventories

4.3 Accounting hand-off

Each transaction produces balanced Dr/Cr lines in RR_INV_TXN_ACCOUNTING using the account rules from section 3.13 and the parameter accounts. The 'Create Accounting' action groups unposted lines by organization and date and posts one journal batch through the existing GL journal interface, writing the batch id back to the transaction (JE_BATCH_ID, ACCOUNTED_FLAG = Y). Journal lines carry REFERENCE2 = TRANSACTION_ID and REFERENCE5 = 'INV_TRANSACTIONS', following the same reference conventions used by AR, AP and Cash, so Check Accounting and the journal drill can link back to the inventory transaction.

5. Inter-Organization Transfers

5.1 Direct transfer (one step)

```
W1 (source)                                W2 (destination)
|  INTERORG_DIRECT: -qty STORES             |
|----- goods move ----->|  INTERORG_DIRECT: +qty STORES

Accounting (source org):  Dr Inter-org Receivable      Cr Material account
Accounting (destination): Dr Material account          Cr Inter-org Payable
Shipment row: TRANSFER_MODE = DIRECT, STATUS = RECEIVED immediately
```

5.2 In-transit transfer (ship, then receive)

```
Step 1  SHIP      source org:      -qty subinventory, + in-transit value
                  Dr In-transit account      Cr Material account (source)
                  Shipment STATUS = SHIPPED

Step 2  RECEIVE   destination org: + qty into subinventory
                  Dr Material account (dest) Cr In-transit account
                  Shipment STATUS = RECEIVED
```

The Inter-Org Transfers page lists shipments with status chips; a SHIPPED row offers the Receive action to the destination organization.

Quantities in transit are visible on the On-Hand Inquiry page as an IN-TRANSIT bucket per destination organization, derived from SHIPPED shipments.

6. Web Services (ORDS)

All endpoints live in the existing reerp module under the inv/ prefix, follow the established JSON conventions (spaced "success": literals, distinct literal paths, no undeclared binds), and return {"success": true/false, ...}. Inquiry screens may also use the guarded SQL gateway (POST ai/executequery) with appUser 'INV_MANAGEMENT'.

Method + Path	Purpose
GET inv/orgs	List organizations (with type, master, BU, status)
POST inv/orgs/create	Create master or child organization
PUT inv/orgs/:organizationId	Update name, location, status
GET inv/orgs/:organizationId/parameters	All three parameter sets in one response
POST inv/orgs/:organizationId/parameters	Upsert inventory parameters
POST inv/orgs/:organizationId/receiving	Upsert receiving parameters
POST inv/orgs/:organizationId/shipping	Upsert shipping parameters
GET inv/orgs/:organizationId/subinventories	List subinventories
POST inv/subinventories/create	Create subinventory
PUT inv/subinventories/:subinventoryId	Update / deactivate
GET inv/subinventories/:subinventoryId/locators	List locators
POST inv/locators/create	Create locator
GET inv/uoms	UOM lookup
GET inv/costorgs	List cost organizations with assigned inv orgs
POST inv/costorgs/create	Create cost organization
POST inv/costorgs/assign	Assign an inventory org to a cost org
GET inv/costorgs/:costOrgId/itemcosts	Item costs in a cost org
POST inv/itemcosts/set	Create or update an item cost
GET inv/items	Search items (by number, description, type, status)
POST inv/items/create	Create item in the master org
PUT inv/items/:itemId	Update item attributes
GET inv/items/:itemId/assignments	Organizations the item is assigned to
POST inv/items/:itemId/assignments	Assign the item to one or more child orgs
POST inv/transactions/create	Create any transaction (type-driven, section 4.1)
GET inv/transactions	Transaction history (org, item, date, type filters)
POST inv/transactions/accounting	Create Accounting for unposted transactions
GET inv/onhand	On-hand by org / subinventory / item (+ in-transit bucket)
POST inv/interorg/ship	Create DIRECT or INTRANSIT shipment with lines
POST inv/interorg/receive	Receive a SHIPPED in-transit shipment
GET inv/interorg/shipments	List shipments with status and lines

6.1 Sample payloads

Create child organization - POST inv/orgs/create

```
{
  "organizationCode": "W1",
  "organizationName": "DIFC Main Warehouse",
  "orgType": "CHILD",
  "masterOrgCode": "MST",
  "businessUnitName": "BUIMERC CORP_DIFC_INVST",
  "location": "DIFC, Dubai",
  "createdBy": "KHALID"
}
-> 201 { "success": true, "organizationId": 5000000012 }
```

Create item - POST inv/items/create

```
{
  "itemNumber": "FG-1001",
  "description": "Steel Bracket 40mm",
  "itemType": "FINISHED_GOOD",
}
```

```

    "primaryUomCode": "EA",
    "listPrice": 25.50,
    "standardCost": 14.00,
    "purchasableFlag": "Y",
    "saleableFlag": "Y",
    "createdBy": "KHALID"
  }
  -> 201 { "success": true, "itemId": 5000000101 }

```

Assign item to organizations - POST inv/items/5000000101/assignments

```

{ "organizationIds": [5000000012, 5000000013], "createdBy": "KHALID" }
-> 201 { "success": true, "assigned": 2, "skippedExisting": 0 }

```

Miscellaneous receipt - POST inv/transactions/create

```

{
  "txnTypeCode": "MISC_RECEIPT",
  "organizationId": 5000000012,
  "itemId": 5000000101,
  "subinventoryCode": "STORES",
  "quantity": 100,
  "transactionDate": "2026-09-19",
  "reason": "Opening stock load",
  "createdBy": "KHALID"
}
-> 201 { "success": true, "transactionId": 5000000501,
        "unitCost": 14.00, "txnValue": 1400.00, "newOnhand": 100 }

```

Subinventory transfer - POST inv/transactions/create

```

{
  "txnTypeCode": "SUBINV_TRANSFER",
  "organizationId": 5000000012,
  "itemId": 5000000101,
  "subinventoryCode": "STORES",
  "transferSubinventory": "FG",
  "quantity": 20,
  "transactionDate": "2026-09-19",
  "createdBy": "KHALID"
}
-> 201 { "success": true, "transactionIds": [5000000502, 5000000503] }

```

In-transit shipment - POST inv/interorg/ship

```

{
  "transferMode": "INTRANSIT",
  "fromOrgId": 5000000012,
  "toOrgId": 5000000013,
  "shipMethod": "Road - Own fleet",
  "lines": [
    { "itemId": 5000000101, "subinventoryCode": "FG", "quantity": 10 }
  ],
  "createdBy": "KHALID"
}
-> 201 { "success": true, "shipmentId": 5000000601,
        "shipmentNumber": "SHP-000001", "status": "SHIPPED" }

```

Receive shipment - POST inv/interorg/receive

```

{ "shipmentId": 5000000601, "subinventoryCode": "STORES", "createdBy": "KHALID" }
-> 200 { "success": true, "status": "RECEIVED", "transactionIds": [5000000604] }

```

7. Screen Designs

All pages live under a new Inventory (INV) module in the menu, using the standard Re-ERP page frame: breadcrumb, title block with module color, search cards, result grids with column sorting and grid search, tabbed detail workspaces, and API-transparency buttons.

7.1 Inventory Organizations

```
+-----+
| Inventory Organizations                                     [API] [Refresh] [+ Create Org] |
+-----+
| Search: [code/name...]  Type: [All v]  Status: [Active v] |
| Code | Name              | Type | Master | Business Unit | Status |
| MST  | BUIMERC Master        | MASTER | -      | -              | Active |
| W1   | DIFC Warehouse       | CHILD | MST    | BUIMERC CORP_DIFC | Active |
| W2   | JAFZA Warehouse      | CHILD | MST    | BUIMERC CORP_FZE  | Active |
+-----+
Row click opens the org workspace as a closable tab (Manage Assets pattern):
[ Parameters | Receiving | Shipping | Subinventories ]
Each tab is a form card with Save; accounts use AccountSelector with the
company segment locked from the BU. Subinventories tab = grid + Add dialog
+ per-row Locators drill when locator controlled.
```

7.2 Item Master

```
+-----+
| Item Master (org: [MST - Master v])                       [API] [Refresh] [+ Create Item] |
+-----+
| Search: [number/description]  Type: [All v]  Status: [Active v] |
| Item      | Description          | Type | UOM | Status | Assigned Orgs |
| FG-1001   | Steel Bracket 40mm   | FG   | EA  | Active | W1, W2 [Assign...] |
+-----+
Create/Edit Item dialog: number, description, type, UOM, category, flags
(stockable / transactable / purchasable / saleable), list price, standard
cost. 'Assign...' opens a checklist of child orgs with default subinventory
per org; saving POSTS the assignments and shows per-org success marks.
```

7.3 Create Transaction

```
+-----+
| Create Transaction                                         [API] [History] |
| Type: [Miscellaneous Receipt v]  Org: [W1 v]  Date: [19-Sep-2026] |
| Item: [FG-1001 (LOV)]  Subinv: [STORES v]  (To Subinv: [FG v] transfers) |
| Qty: [100]  UOM: EA  Unit Cost: 14.00 (from cost org, read-only) |
| Reason: [Opening stock load] |
| [ Preview accounting ]  [ Create ] |
| Preview shows the Dr/Cr lines that will be generated before committing. |
+-----+
```

7.4 Inter-Org Transfers

```
+-----+
| Inter-Org Transfers                                       [API] [Refresh] [+ New Transfer] |
| Shipment | From | To | Mode      | Status | Date      | Actions |
| SHP-00001 | W1   | W2 | INTRANSIT | SHIPPED | 19-Sep-26 | [Receive] [View] |
| SHP-00002 | W1   | W3 | DIRECT    | RECEIVED | 18-Sep-26 | [View] |
+-----+
New Transfer wizard: 1) From org + To org + mode 2) Lines (item LOV limited
to items assigned in BOTH orgs, subinv, qty, available shown) 3) Review ->
Ship. Receive dialog asks destination subinventory per line.
```

7.5 On-Hand Inquiry

```
+-----+
| On-Hand Inquiry      Org: [W1 v]  Subinv: [All v]  Item: [search...] |
| Item      | Description          | Subinv | Locator | On-Hand | UOM | Value |
| FG-1001   | Steel Bracket 40mm   | STORES | -        | 80      | EA  | 1,120.00 |
| FG-1001   | Steel Bracket 40mm   | FG     | -        | 20      | EA  | 280.00 |
| IN-TRANSIT to W2 | | | | 10      | EA  | 140.00 |
+-----+
Runs on the guarded SQL gateway; Excel export; drill from a row to the
transaction history filtered to that item/subinventory.
```

7.6 Cost Organizations

```

+-----+
| Cost Organizations      [API] [Refresh] [+ Create Cost Org] |
| Code | Name           | Method | Currency | Assigned Inv Orgs | Status |
| CO-01 | BUIMERC Costing| AVERAGE | AED      | W1, W2 [Assign...] | Active |
+-----+

```

Tabs per cost org: Assignments (add/remove inv orgs) and Item Costs (grid of item / cost / effective date with inline set-cost dialog).

8. Numbering, Seeding and Conventions

- RR_INV_SEQ starts at 5000000001 and feeds every PK in the module
- Receipt and shipment numbers come from the per-org counters in the parameter tables
- Seed script loads RR_INV_UOMS (10 common units) and RR_INV_TXN_TYPES (section 3.13)
- All handlers follow the module's hard-won ORDS rules: distinct literal paths, spaced "success": literals, ORDS.DEFINE_PARAMETER for any GET parameter, EXECUTE IMMEDIATE for optional-table access, one script per feature numbered database/inv/2xx_*.sql
- Journal references: REFERENCE2 = TRANSACTION_ID, REFERENCE5 = 'INV_TRANSACTIONS'

9. Delivery Roadmap

Phase	Content	Depends on
1a - Setup	Orgs + parameters + subinventories + cost orgs + UOMs (tables, services, pages)	-
1b - Items	Item master + org assignments + item costs	1a
1c - Transactions	Transaction engine + on-hand + misc receipt/issue + subinv transfer + Create Accounting	1b
1d - Inter-org	Direct and in-transit transfers + on-hand in-transit bucket	1c
2 - Purchasing	Suppliers linkage, requisitions, purchase orders, PO receiving into the same transaction engine	1c
3 - Order Management	Sales orders, reservations, pick and ship using SO_ISSUE transactions	1c
4 - Cost Accounting	Perpetual average recalculation, cost adjustments, valuation reports per cost org	1c

Each sub-phase ships as one database script plus its pages, deployable independently - the same delivery style used for the Banks and FA features.

10. Open Decisions

- Should the master organization also be transactable, or setup-only? (Recommendation: setup-only)
- One cost organization per company to start, or allow several from day one? (Schema supports both)
- Should Create Accounting run automatically after every transaction, or stay a manual/periodic action like the rest of Re-ERP? (Recommendation: manual button + optional daily run later)
- Locators: enable in Phase 1 UI or keep schema-only until a warehouse needs them?